srt-router user guide

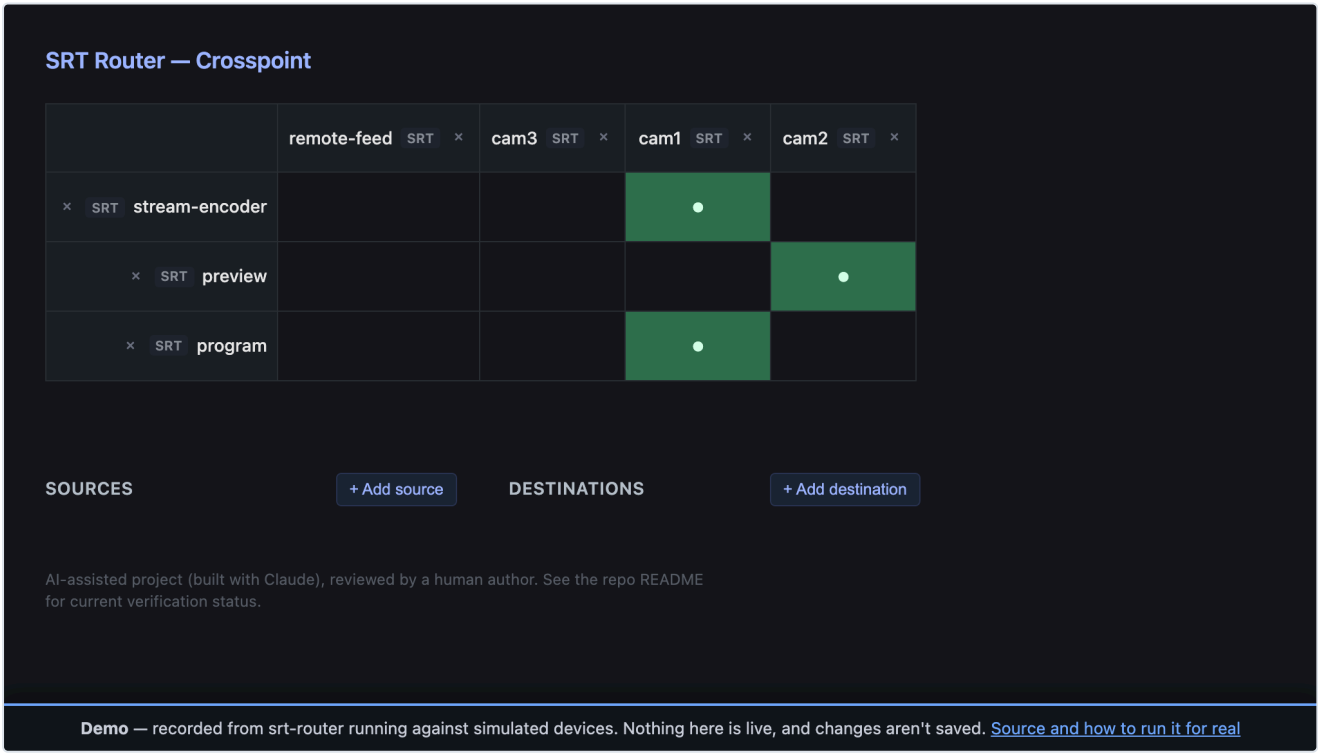
srt-router is a broadcast video router for SRT streams. If you've used an SDI or HDMI router, you already know how it works: a grid of sources down one side and destinations along the other, where each destination is fed by exactly one source, switchable live.

> **Before you rely on this:** I've run the relay engine and web UI locally, including integration tests relaying real SRT and NDI traffic end to end and a live crosspoint switch from the UI. But it's **never been run against real third-party encoders or decoders, or over a real (non-loopback) network path**. Check it over before you put it in a live chain.

The mental model

- A **source** is a stream coming in (a camera encoder, a remote feed, a still image).
- An **output** is a stream going out (to a decoder, a streaming platform, a monitor).
- The **crosspoint** connects them: **each output takes exactly one source; one source can feed any number of outputs.**

Switching an output from one source to another is what it's for, and it happens live.



One marked cell per row, because an output takes exactly one source. A column with several marks is one source feeding several outputs, which is allowed and normal.

Running it

```
srtrouter --config /path/to/config.toml
```

Then open the web UI at whatever `[web] bind` says (`http://localhost:8080` with the example config). You'll get the crosspoint grid, and it updates live: if someone else switches an output, your screen follows straight away.

Setting it up

Configuration is a single TOML file. A minimal working router:

```
[web]
bind = "0.0.0.0:8080"

[[inputs]]
id = "cam1"
transport = "srt"
mode = "listener"      # we wait; the encoder connects to us
bind = "0.0.0.0:5001"

[[inputs]]
id = "cam2"
transport = "srt"
mode = "listener"
bind = "0.0.0.0:5002"

[[outputs]]
id = "program"
transport = "srt"
mode = "listener"      # the decoder connects to us
bind = "0.0.0.0:6001"
default_source = "cam1"
```

listener or caller?

This trips people up more than anything else.

- **`listener`**: srt-router *waits*, and the other end connects to it.
- **`caller`**: srt-router *dials out* to an address you give it.

Every SRT link needs exactly one of each. If both ends are listeners, nothing ever connects and both sides sit there waiting for the phone to ring. Two callers is the same problem the other way

round: both ringing, nobody picking up. The usual arrangement is encoders **calling in** to the router's input ports, and the router **calling out** to decoders, but any combination works as long as each link has one of each.

Do you want routing to survive a restart?

```
[state]
path = "state/routes.json"
```

- **With `[state]`**: switches you make in the UI are saved and **restored on restart**, overriding every `default_source`.
- **Without it**: the router **returns to the `default_source` values every time it starts**.

Both make sense. Persisting is what you want for an installation, and resetting is what you want for a truck that should come up in a known state every show day.

Beyond SRT

Transports only mix where the payload matches. SRT and media both carry plain MPEG-TS, so a still image can feed an SRT output. NDI only routes to NDI and OMT only to OMT, because the router doesn't transcode between them.

Transport	Use	Availability
SRT	The main event	Always
NDI	Local-network video	Built in (needs the NDI runtime when it runs)
OMT	Open Media Transport	Only if your build has it
media	Stills, file playback, rescaling	Always (needs `ffmpeg` installed)

Check what your build supports: visit `/api/manage/transports` or look at the UI's add menu. NDI is in every release and in a default build from source, and loads the NDI runtime when it runs. Without the runtime, an NDI endpoint fails and the message names the download. OMT needs a build with the `omt` feature and the OMT SDK. `media` needs only `ffmpeg` on your `PATH`.

Useful media tricks

```
[[inputs]]                    # a slate to cut to when a feed dies
id = "slate"
transport = "media"
mode = "stills"
image_path = "/opt/srtrouter/slate.png"

[[inputs]]                    # a downscaled version of an existing source
id = "cam1-720p"
transport = "media"
mode = "scaler"
source = "cam1"
width = 1280
height = 720
```

Media publishes plain MPEG-TS, the same format SRT relays, so a slate feeds an SRT output with **no transcoding step**.

Note media is **input-only**. There's no such thing as a media output.

Adding and removing things while running

You don't have to restart to add a source or output. The UI can do it live, and anything added this way behaves the same as something declared in the config file. Removing an output does free its port.

The one catch: **things added at runtime aren't written back into your config file.** If you want them next time, add them to the TOML too (or use `[state]`, which persists *routing* but not the set of sources and outputs).

Troubleshooting

A source never connects. Almost always a listener/caller mismatch, so check both ends. Then check the port isn't firewalled, and that any SRT stream ID or passphrase matches.

"can't route a X source to a Y output without transcoding". The two payload kinds are incompatible, and the router is refusing to emit something the far end couldn't decode. SRT and media sources can feed SRT outputs; NDI and OMT only route to their own kind. A `media` `scaler` only rescales MPEG-TS, so it can't convert NDI or OMT either.

NDI or OMT isn't in the menu. Your build doesn't have it. Check `/api/manage/transports`. NDI is in every release and in a default build, so a missing NDI means a build made with `--no-default-features`. OMT needs a build with the `omt` feature and the OMT SDK.

Routing resets every time I restart. You have no `[state]` section. Add one.

Adding something says it's a conflict. That `id` is already in use. Ids must be unique across sources, and across outputs.

I want more detail in the logs.

```
RUST_LOG=debug srtrouter --config config.toml
```

Security

There's no authentication on the web UI or API. Anyone who can reach the port can re-route your program output or delete a source. Put it on a management network, bind it to a specific interface, or firewall it, and don't expose it to the internet.
